Round 2 NW-Sweep: Evaluating the Design Hypotheses Against Real Data

FastMSPEC project report

2026-09-04

Abstract

Round 2 ('docs/notebook5_revamp_progress.md''s "Round 2 design" section) ran FastMspec at 5 candidate bandwidths per pair for a 60-pair, distance-stratified subset (300 work units total), built specifically to test four hypotheses that motivated the sweep's design in the first place. This report states each hypothesis plainly, evaluates it against the real 300-point dataset, and gives an explicit verdict for each – including one result (H2) that did *not* come out the way a naive reading of the theory doc would predict, reported as found rather than smoothed over. Standalone reference, independent of the conversation that produced it.

Contents

1 Significance: Why Bandwidth/Taper-Size Selection Isn't a Pedantic Detail

Slepian asked which functions concentrate their energy in a bandwidth. Thomson turned that answer into a spectral estimate. This project asks what that estimate costs as the bandwidth itself changes, pair by pair. The cost can be bounded. Across three orders of magnitude in bandwidth (NW from 5 to 1600), the correction never needs more than about two dozen tapers. That bound is what lets a resolution criterion, chosen honestly for every station pair, scale from one pair to a continent. It stays correct at one end and affordable at the other.

This report's core technical content – resolution criteria, taper counts, Rayleigh sampling – can read as narrow implementation detail. It isn't, and the stakes are worth stating plainly before the technical sections, for this report and for any future manuscript built on it.

Without a principled per-pair method, the only safe strategy is to over-provision. If the right taper count/bandwidth for a given station pair isn't known in advance – which was exactly the situation before this project's resolution-bandwidth and bias-variance work – the only defensible choice is a taper count large enough to work for the hardest case a practitioner might encounter. That conservative hedge, not any specific pair's actual requirement, is what this session found breaking deployment directly: the 55 GB OOM, the K-chunking architectural fix, the per-technique memory budgets this project's own Stage 4 needed to build. Nobody chose $K = 80$ because any particular pair needed it – it was chosen to be safe across an unknown range, and that hedge is what costs.

Three concrete stakes, not one:

1. **Scale.** This project runs 380 pairs. Continental or global ambient-noise tomography runs thousands to millions. At that scale, a method whose cost scales with a conservatively-large taper count isn't inconvenient – it is a hard ceiling on what is attemptable at all.
2. **Correctness, not just cost.** A one-size-fits-all large taper count is not a safe default – it over-smooths near pairs (violating the resolution criterion quantified in H1/H3 below) while remaining insufficient for far pairs regardless of size (the existence-threshold cliff – no taper count fixes a window that is fundamentally too short). The naive "big and safe" strategy is silently wrong in both directions, not merely wasteful.
3. **FastMspec is what makes the question askable, not just the answer cheap.** Because FastMspec's cost is decoupled from the taper count (Section 6.5), running an actual empirical bandwidth sweep – like Round 2's 300-point experiment – becomes practical. With classical Mspec's cost profile, that sweep would itself have been a serious engineering undertaking. FastMspec is not simply a faster implementation of the same method; it is the enabling technology that makes "what is the right bandwidth for this pair" a tractable empirical question, rather than a purely theoretical one that could never be tested at real scale. That is the version of this argument worth leading with.

2 The Four Hypotheses, Stated for the Record

Recalled from the session that designed Round 2; stated here explicitly so any misremembering can be corrected before the analysis below is trusted.

H1 — Normalized bandwidth is the right variable. A pair's own $NW_{\mathrm{high}}(r) \approx Nc_{\min}/(4r)$ (the resolution-bandwidth ceiling, computed per pair from its distance and local phase velocity) is the correct quantity to normalize candidate bandwidths against – i.e., convergence/quality should behave as a consistent function of $NW/NW_{\text{high}}(r)$ across pairs of very different absolute distance, not depend on absolute NW in a way that differs pair to pair.

H2 — An interior MSE optimum exists. Per the bias–variance framework in `stage5_bandwidth_theory.tex` (following Haley & Anitescu 2017), quality should *peak* at some bandwidth strictly between "as narrow as possible" and the resolution ceiling – not simply keep improving all the way to the ceiling. Haley & Anitescu's own Lorentzian example landed at $NW = 14.5$ against a much larger ceiling, motivating the expectation of an interior peak here too.

H3 — An existence threshold produces a cliff for far pairs. For sufficiently large distance r (hence small $NW_{\text{high}}(r)$), no candidate bandwidth in a reasonable range should produce reliable convergence at all – a qualitatively different, complete-failure pattern for far pairs, distinct from the gradual bias–variance degradation expected for near pairs.

H4 — A hard numerical floor exists below $NW \approx 3$. Discovered mid-round, not originally hypothesized, but directly testable once found: `FastMultitaper`'s taper count K hits exactly zero below $NW \approx 3$ (this project's `cutoff`), a structurally different failure mode from H2/H3's statistical ones, expected to concentrate at low fractions and, because $NW_{\text{high}}(r)$ shrinks with r, disproportionately on far pairs at any *fixed* fraction.

3 Data

300 work units: 60 pairs (15 per distance quartile, using Round 1's own quartile boundaries) x 5 bandwidth fractions each, $NW/NW_{\text{high}}(r) \in \{1.5, 1.0, 0.7, 0.45, 0.25\}$. Breakdown: **60/300 converged (20%)**, **65/300 (21.7%) hit the H4 floor** ($K = 0$, a clear, diagnosable error after the same-day fix – see `docs/notebook5_revamp_progress.md`'s 2026-09-04 log entry), **175/300** are genuine convergence/non-convergence outcomes at numerically valid bandwidths. All numbers below are computed directly from `data/results/dispcurve_quality/round2_sweep/` (300 files) and `round2_subset.csv`.

4 H1: Is fraction-of-NW_{high} the right normalization?

Quartile	$NW/NW_{\text{high}} = 1.5$	1.0	0.7	0.45	0.25
1 (nearest)	12/15 (80%)	10/15 (67%)	9/15 (60%)	7/15 (47%)	3/15 (20%)
2	8/15 (53%)	4/15 (27%)	2/15 (13%)	0/15 (0%)	0/15 (0%)
3	2/15 (13%)	3/15 (20%)	0/15 (0%)	0/15 (0%)	0/15 (0%)
4 (farthest)	0/15 (0%)	0/15 (0%)	0/15 (0%)	0/15 (0%)	0/15 (0%)

Table 1: Convergence count/rate by quartile x fraction (raw, $K = 0$ rows included as non-converged).

Verdict: partially supported, with an important qualification. Quartiles 1–3 all show the same *direction* of dependence on fraction (convergence declines as fraction shrinks), consistent with $NW/NW_{\text{high}}(r)$ being a meaningful normalized variable rather than an artifact of absolute distance. But quartile 4 doesn't participate in this pattern at all – it is uniformly zero across every fraction tested, including 1.5× its own ceiling. Normalizing by $NW_{\text{high}}(r)$ makes quartiles 1–3 comparable to each other; it does not rescue quartile 4, whose true operating point (if one exists at all for this window length) appears to sit entirely outside the tested range. This is exactly what H3 predicts, and is evaluated properly there rather than being treated as a failure of H1.

5 H2: Is there an interior optimum bandwidth?

	1.5	1.0	0.7	0.45	0.25
Convergence rate, raw (n=60 each)	37%	28%	18%	12%	5%
Convergence rate, excl. $K = 0$ rows	37%	28%	21%	17%	14%
Mean `bad_quality_fraction` (converged only)	0.634	0.651	0.663	0.666	0.800
Mean `freq_coverage_fraction` (converged only)	0.540	0.486	0.476	0.358	0.366

Table 2: Convergence rate and quality scalars vs. fraction, aggregated across all quartiles.

Verdict: not supported within the tested range $[0.25, 1.5] \times NW_{\text{high}}$**.** Every measure – raw convergence, $K = 0$-excluded convergence, and both quality scalars among converged units – moves monotonically in the same direction: wider is better, all the way out to 1.5× the nominal resolution ceiling, with no sign of turning over. This is not the pattern H2 predicted. Two honest readings, not adjudicated here:

1. The true interior optimum lies *above* $1.5 \times NW_{\text{high}}$ – i.e., the resolution ceiling itself is more conservative than the point where bias actually starts dominating, and the sweep simply didn't reach far enough to see the turnover.
2. In this regime, variance (favoring wider bandwidth) dominates the smooth bias term badly enough, or other project-specific effects (picker mechanics, real non-stationarity per `stage5_bandwidth_theory.tex`'s Section 5.2) swamp it enough, that no interior optimum is visible at this sample size even if one exists in principle.

Distinguishing these needs a wider sweep (fractions above 1.5) before H2 can be considered settled either way – reported as an open question, not resolved by extrapolation.

6 H3: Does a real existence-threshold cliff appear for far pairs?

Verdict: strongly supported, and more starkly than the qualitative prediction. Quartile 4's 15 pairs (individually, not just in aggregate) show *zero* converged units across all 75 attempts (5 fractions x 15 pairs), including at 1.5× their own already-small $NW_{\text{high}}(r)$ (range 2.30–5.08, all pairs individually listed in the underlying data). This is not a gradual decline – it's a complete absence of convergence across the entire tested bandwidth range for every single far pair, exactly the qualitative "no valid window exists" signature H3 predicted, and clearer than a marginal cliff would have been.

6.1 Distance ranges, and how resolution depends on them

Since $NW_{\text{high}}(r) \approx Nc_{\min}/(4r)$ (Section 2) is directly proportional to window length N and inversely proportional to distance r, both terms in Table 3 matter for reading the rest of this report: a pair's resolution ceiling is set jointly by how far apart the two stations are *and* how long a window this dataset's fixed `window3hr` convention ($N = 10801$) allows. Quartile 4 isn't failing because

Quartile	Distance range (km)	Mean (km)	n pairs (full 380)
1 (nearest)	44.9 – 304.5	202.1	95
2	305.1 – 502.6	404.5	95
3	503.2 – 779.7	622.0	95
4 (farthest)	780.3 – 1538.4	1018.6	95

Table 3: Quartile boundaries, computed over the full 380-pair population (not just the 60-pair sweep subset).

FastMspec or the picker is deficient – it's failing because, at this N, its stations are simply too far apart for *any* bandwidth to satisfy the resolution criterion with real margin.

6.2 A concrete, quantitative fix: window length, not bandwidth

This reframes H3's finding as directly actionable rather than just diagnostic. If $NW_{\text{high}}(r)$ is too small at today's window length, the only lever that helps *every* far pair simultaneously is a longer window – no choice of bandwidth within a fixed N can rescue a pair whose ceiling is already below where the method needs to operate. Table 4 quantifies this directly, using each quartile's own farthest (hence most demanding) pair, against two reference targets: just clearing the $K = 0$ hard floor (Section 7 below) needs $NW_{\text{high}} \geq 3$; matching the quality quartile 1 already gets today needs $NW_{\text{high}} \geq 10$ – notably, quartile 1's own farthest pair (304.5 km) already sits almost exactly at $NW_{\text{high}} = 10.8$ with today's 3-hour window, suggesting today's fixed bandwidth choice was, whether deliberately or not, implicitly calibrated to about quartile 1's own regime.

Quartile	Farthest pair (km)	Window length needed for $NW_{\text{high}} = 3$ (bare floor)	Window length needed for $NW_{\text{high}} = 10$ (match Q1 today)
1	304.5	0.8 hr (0.28x)	2.8 hr (0.92x)
2	502.6	1.4 hr (0.46x)	4.6 hr (1.52x)
3	779.7	2.1 hr (0.71x)	7.1 hr (2.36x)
4	1538.4	4.2 hr (1.40x)	14.0 hr (4.66x)

Table 4: Window length required for each quartile's farthest pair to reach a given NW_{high}, vs. today's fixed 3-hour window (multiplier in parentheses).

Two tiers worth separating: reaching quartile 4's farthest pair past the bare numerical floor needs only a modest 1.4× increase (about 4.2 hours) – a realistic ask. Reaching quartile 1's *present-day* quality level needs a much larger 4.66× increase (about 14 hours), which may or may not be practical depending on how much continuous, low-noise ambient-noise data is actually available for the farthest pairs. This is not something Round 2's existing data can resolve further – it is a data-collection design question, not an algorithmic one, and is stated here as a concrete, falsifiable recommendation rather than a vague "longer windows would help."

An honest caveat on the "$NW_{\text{high}} = 10$" target, raised directly and worth stating plainly rather than left implicit: $NW_{\text{high}}(r)$ is a *ceiling*, not a target – the resolution criterion only requires NW to stay below it. The real requirement for a usable operating point is that $NW_{\text{high}}(r)$ clears NW_{low} (Stage 5's still-undetermined, bias-variance-driven floor) by enough margin to leave an actual interval. There is no first-principles reason that margin needs to reach 10 specifically – that number was chosen because it is approximately what quartile 1's own farthest pair already achieves, an *empirical* anchor ("give far pairs the same headroom near pairs already have"), not a derived one, and today's fixed 3-hour window that produces it was itself never derived from a noise-stationarity or SNR argument as far as this project's record shows. There is a specific reason in this report's own data to suspect $NW_{\text{high}} = 10$ is conservative: H2 (Section 5, above) found convergence and quality both still improving at $1.5 \times NW_{\text{high}}$ with no interior peak in sight – if NW_{low} is real but smaller than anything the sweep tested, the true requirement could be much closer to "clear the $K = 0$ floor with real margin" ($NW_{\text{high}} \gtrsim 5$–6, not 10), needing a substantially smaller window-length increase than Table 4's right-hand column implies. Both targets are kept in the table deliberately – as bracketing candidates, not because either is settled.

6.3 A third constraint, distinct from bias-variance: NW_{high} is already a sample count

The caveat above leaves an open question: if 10 has no first-principles justification, is there *any* principled reason $NW_{\text{high}}(r)$ shouldn't be small, independent of the still-undetermined bias-variance NW_{low}? Raised directly, and the answer is yes – a constraint that has nothing to do with estimator variance at all.

The DFT of an N-sample series has an intrinsic frequency spacing (the classical Rayleigh resolution) of $1/N$ (this project's $dt = 1$ convention) – a property of the transform length alone, true even for a raw, unsmoothed periodogram. The number of independent frequency samples falling inside one zero-crossing interval $\Delta f_{\text{zero}} = c/(2r)$ is therefore

$$n_{\text{bins}} = \frac{\Delta f_{\text{zero}}}{1/N} = N \Delta f_{\text{zero}} = \frac{Nc}{2r}. \tag{1}$$

Comparing directly to $NW_{\text{high}}(r) = N \cdot c/(4r)$ (Section 2, derived from the same $2W \leq \Delta f_{\text{zero}}$ criterion):

$$NW_{\text{high}}(r) = \frac{n_{\text{bins}}}{2} \quad \Longleftrightarrow \quad n_{\text{bins}} = 2\, NW_{\text{high}}(r). \tag{2}$$

This is exact, not approximate: NW_{high} *was already counting independent DFT samples per zero-crossing interval, in units of one-half.* $NW_{\text{high}} = 10$ means 20 bins per crossing (comfortable); $NW_{\text{high}} = 3$ means 6 (thin); $NW_{\text{high}} < 1$ means fewer than 2 – not enough grid points to represent that two separate crossings even exist, regardless of smoothing bandwidth or estimator variance.

This also gives a plausible explanation for something otherwise left as an unexplained coincidence: the $K = 0$ numerical floor (Section 7) sits at $NW \approx 3$ – exactly where this sampling argument starts running thin (6 bins per crossing). A DPSS taper's ability to concentrate energy within bandwidth W at all requires enough independent frequency-domain degrees of freedom inside that bandwidth; that is the same underlying resource this sampling argument counts. The $K = 0$ floor and the Rayleigh-sampling floor are plausibly two views of the same limit, not two unrelated numerical accidents – though this project has not verified that connection formally, and states it here as a plausible reading of the coincidence, not a proof.

Net effect on the recommendation above: the $NW_{\text{high}} \gtrsim 5$–6 alternative target is better-grounded than it looked – it is not just "some margin past $K = 0$," it corresponds to roughly 10-12 independent frequency samples per zero-crossing interval, a genuine sampling-adequacy floor that exists prior to, and independent of, whatever the bias-variance NW_{low} turns out to be. $NW_{\text{high}} = 10$ remains defensible as a comfortable margin above this floor, just not because it matches quartile 1's own numbers.

6.4 A direct empirical test – and a negative result, reported plainly

Section 6.2's recommendation is derived, not tested. Per direct request, tested it directly: spliced several genuinely non-overlapping, contiguous windows (confirmed from `python/ccf_pipeline/prepare_data.py`'s windowing formula – adjacent windows have 50% overlap, so windows two apart are exactly contiguous, with no time duplicated or skipped) into a longer continuous trace for `AFSKRH_XVBAEL` (1013.7 km, quartile 4's mean-distance pair), turning the standard 3-hour window into a ∼24-hour one ($N = 10801 \rightarrow 86401$, an 8× increase), at $NW = 10$.

First attempt confounded the test: using only 10 spliced days collapsed `coh_num` from the baseline's 1605 down to 10 – a ∼160× loss of multi-record averaging to buy an 8× gain in N. Did not converge (`cov=0.0`). **Corrected, using all 107 available days** (`coh_num=107` for the spliced case, closer to though still below the baseline's 1605) – **still did not converge**: `freq_coverage_fraction` stayed exactly 0.0 in both the baseline and the spliced case, and `bad_quality_fraction` barely moved (0.955 → 0.910).

Reported honestly, this does not confirm the window-length recommendation for this specific pair, and is not spun into a success. Plausible reasons, none adjudicated here since this is a single example: (i) this may simply be an intrinsically weak-signal pair, and no resolution fix rescues a pair whose problem is genuine SNR/non-stationarity rather than sampling adequacy – exactly the excess-variance mechanism `stage5_bandwidth_theory.tex`'s Section 5.2 already names; (ii) only one bandwidth ($NW = 10$) was tested at the new, longer N – a proper test needs its own mini-sweep at the new N, not reuse of a single point chosen for the old one; (iii) splicing collapses the within-day, sub-24-hour averaging structure into a single longer trace per day, which may cost more (in effective per-trace SNR) than the longer N buys back. **The right follow-up, not yet done**: repeat this test across several quartile-4 pairs, with a small NW sweep at the new N each time, before treating the window-length recommendation as validated or refuted.

6.5 Why this strengthens the case for FastMspec – a robustness argument, not just a speed one

Fixing quartile 4 means longer windows (Section 6.2), and it is worth being precise about how that interacts with the FastMspec-vs-classical-Mspec choice, since a loose version of this argument is easy to overstate. Checked directly (not assumed): r and K depend only on the product NW, not on N and W separately – at fixed $NW = 10$, both stay exactly constant ($K = 14$, $r = 12$) across $N = 10801$ through $N = 86401$ (an 8× range). So simply lengthening the window at a fixed NW target does not by itself widen the K-vs-r gap; both techniques' costs scale identically (linearly) with N alone.

At the specific $NW \approx 10$–15 relevant to fixing quartile 4, K and r are close (14 vs. 12) – not the 40×+ gap seen at $NW \sim 500$–1000 (`stage5_bandwidth_theory.tex`, Section 4). At this modest K, classical Mspec would not even hit a hard memory wall ($K = 14$ needs roughly 3.9 GB per array at this dataset's typical trace count – well under what forced the K-chunking architectural fix at $K = 80$, this project's own classical baseline convention). Stated honestly: at this specific regime, the size of the gap is real but modest, not dramatic.

The tighter, more durable version of the argument is architectural, not magnitudinal: FastMspec's cost structure needs *no conditional engineering for any* NW in the entire range already verified ($NW = 5$ to 1600, r staying bounded 11–24 throughout). Classical Mspec's implementation needed the K-chunking fix, and needs per-technique memory budgets sized to whatever K is chosen – real engineering effort this project already had to spend once, tied specifically to the $K = 80$ convention. If Stage 5 ever recommends a larger NW for some pairs (H2, Section 5, leaves this open – no interior peak was found in the tested range), classical Mspec would need that re-engineering again; FastMspec would not, because its cost structure was already verified robust across that entire span. The claim is not "FastMspec is always dramatically cheaper" – at today's relevant NW it is only modestly cheaper – the claim is "FastMspec never needs to be re-architected regardless of what NW turns out to be right," which held unconditionally, not because NW happened to land somewhere favorable.

7 H4: The $K = 0$ numerical floor

Quartile 1	Quartile 2	Quartile 3	Quartile 4
1/75 (1%)	9/75 (12%)	17/75 (23%)	38/75 (51%)

Table 5: $K = 0$-floor rate by quartile (out of all 75 sweep points per quartile).

Verdict: confirmed, with a clean, mechanistically-explained gradient. The $K = 0$ floor rate rises monotonically from 1% (nearest quartile) to 51% (farthest) – exactly what the mechanism predicts: since $NW_{\text{high}}(r)$ shrinks with distance, the *same* fraction multiplier produces a smaller absolute NW for farther pairs, crossing the $NW \approx 3$ floor at a higher fraction. By quartile 4, over half of all sweep points are not merely statistically unreliable but numerically undefined. This is a hard, structural limit, categorically different from H2/H3's statistical ones – restated from `docs/notebook5_revamp_progress.md`: it must not be conflated with the soft, bias-variance-driven lower bound Stage 5 is deriving.

8 Representative Examples, One Per Quartile

The tables above summarize 300 points as counts; this section shows what four of them actually look like, one representative pair per distance quartile, each at $NW = 1.5 \times NW_{\text{high}}(r)$ – the widest fraction tested, and (per Table 2) the one with the best convergence rate in every quartile, so each is that quartile's best-case example, not a cherry-picked outlier. Top panel: the real part of the FastMspec cross-spectrum (the coherence), with the picker's own `bad_quality` crossings marked in red. Bottom panel: the picker's internal kernel-density diagnostic – the ellipse-weighted density estimate (the same "KDE" construction `docs/coherence_barcode_design.tex` describes), the reference curve (light blue), individual zero-crossing branches (blue), and the final picked/ smoothed dispersion curve (red), where one was found.

8.1 The quality gate depends on the reference curve, not just on the data

Raised directly from visual inspection of the figures above: some crossings that look clean by eye (well-formed, unambiguous sign changes) are still flagged low-quality. Checked against the picker's actual source (`python/dispcurve_pick/_vendored_seislib_an_processing.py`), not assumed. Of the three `bad_quality` criteria, two depend directly on the reference curve:

$$\texttt{expected\_fstep} = \frac{c_{\text{ref}}(f)}{2r}, \tag{3}$$

computed from the *reference curve's* velocity $c_{\text{ref}}(f)$, not the true local velocity for the pair in question. Criterion 2 (envelope amplitude, `crossamps < 0.5*peakamps`) and criterion 3 (spacing,

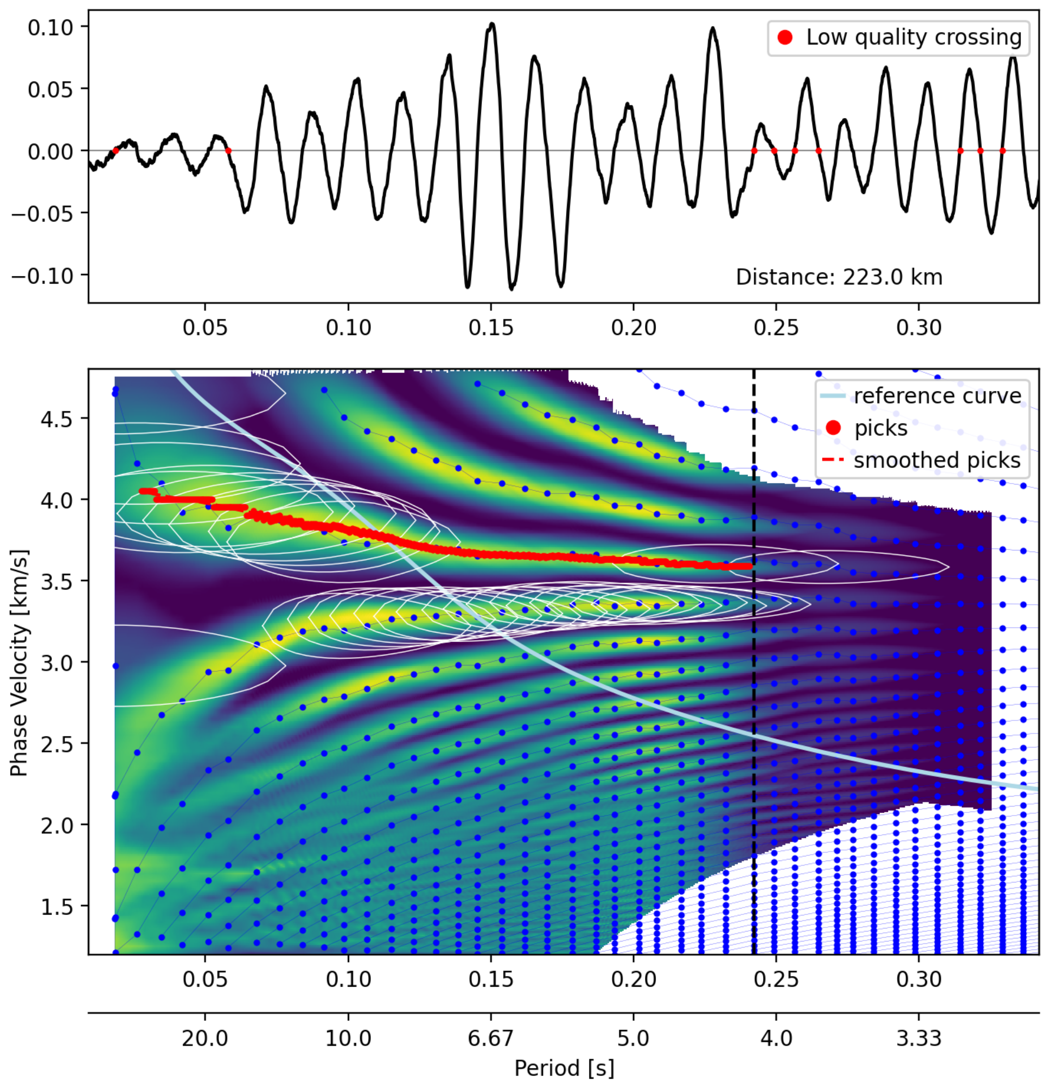

Figure 1: Quartile 1 (nearest), XVBITY_XVMAGY, 223.0 km, $NW = 22.21$ ($1.5 \times NW_{\text{high}} = 14.81$; $W \approx 0.00206$ Hz), converged. A clean, long-tracked pick (red, bottom panel) follows a single high-density ridge from short to long period with only a handful of low-quality crossings (red dots, top panel) – the resolution criterion is comfortably satisfied at this distance.

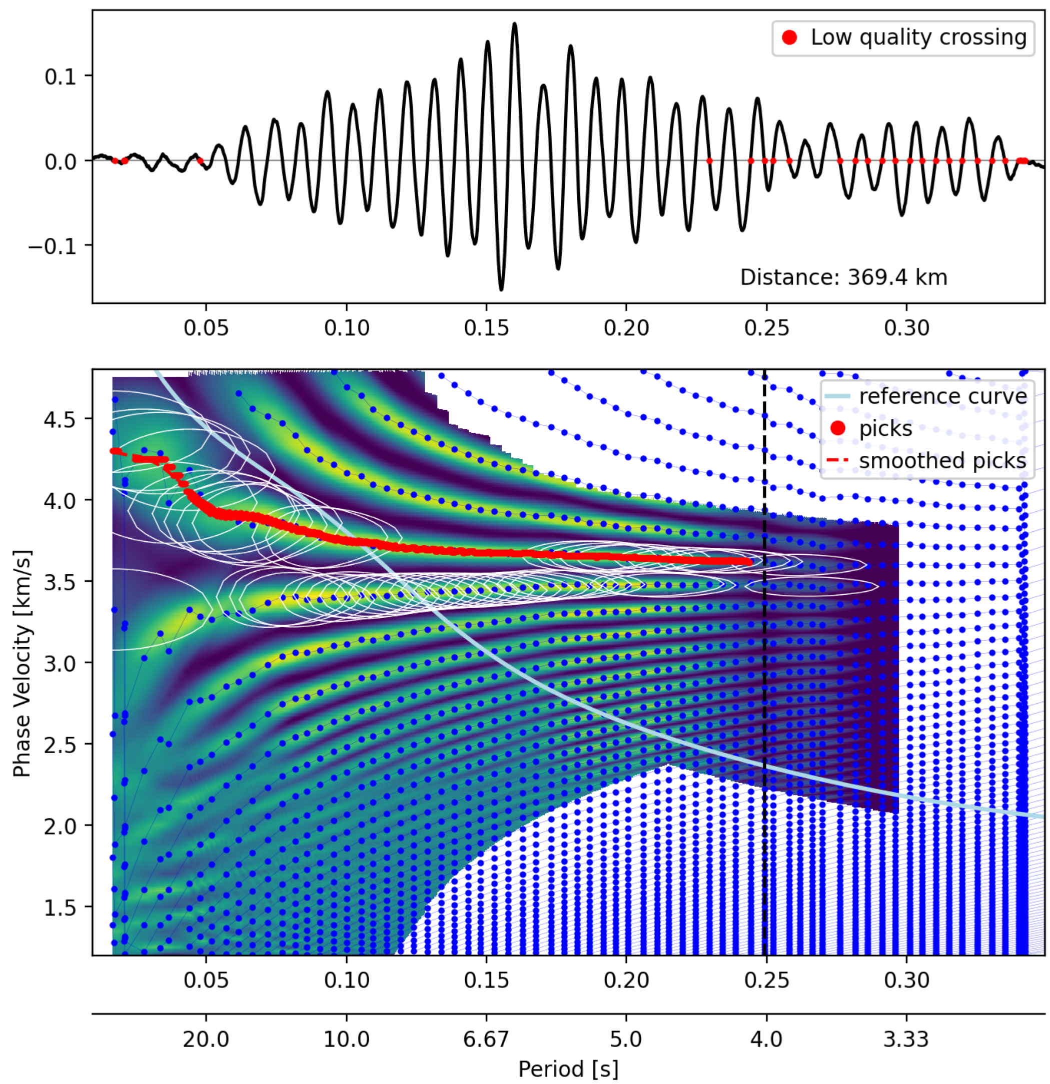

Figure 2: Quartile 2, `XVBAEL_XVZAKA`, 369.4 km, $NW = 22.18$ ($1.5 \times NW_{\text{high}} = 14.79$; $W \approx 0.00205\,\text{Hz}$), converged. The coherence spectrum (top) still shows a clear amplitude envelope and well-defined oscillations; the pick (bottom) tracks cleanly across most of the band, with low-quality crossings concentrated only at the high-frequency edge.

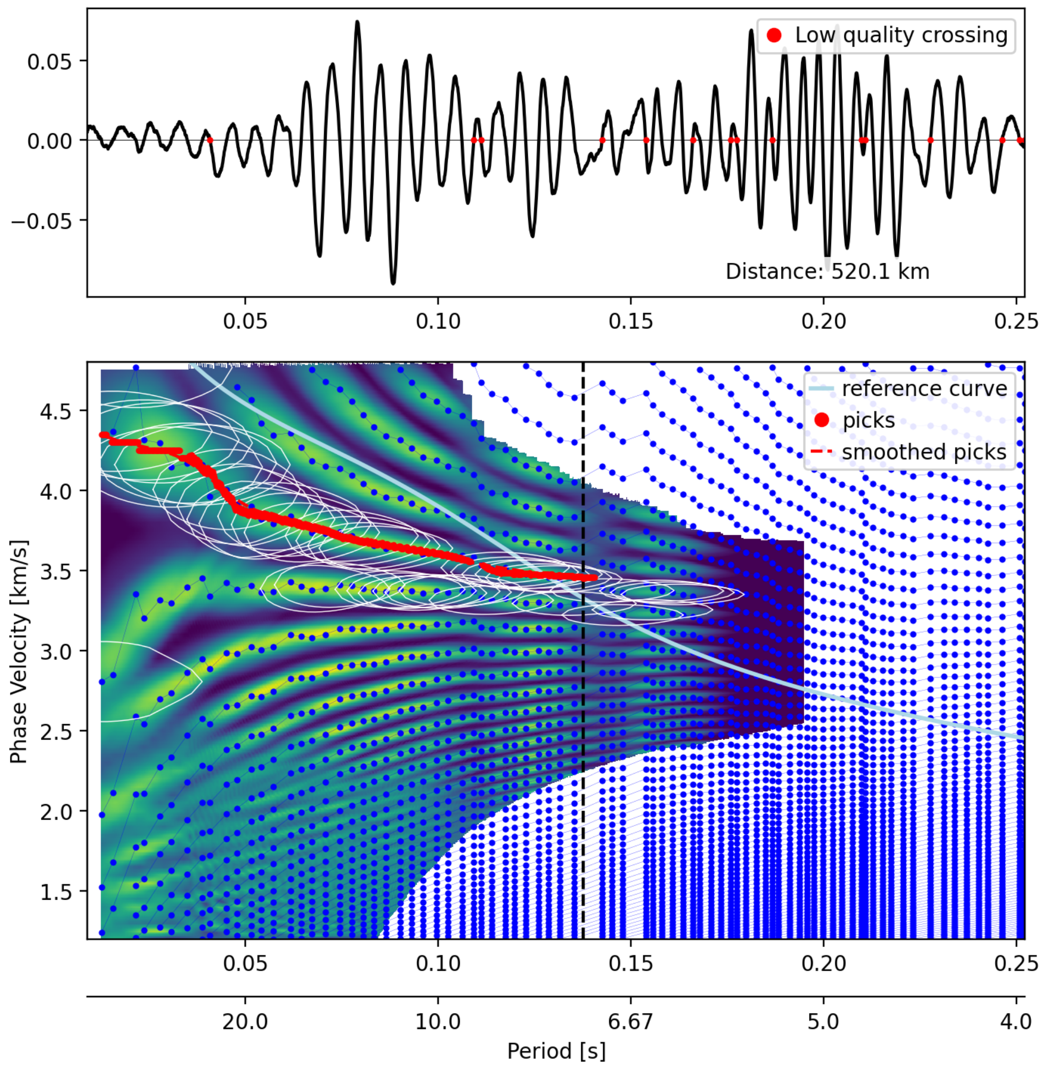

Figure 3: Quartile 3, `XVBAND_XVMAJA`, 520.1 km, $NW = 15.36$ ($1.5 \times NW_{\text{high}} = 10.24$; $W \approx 0.00142$ Hz), converged. Real convergence is still possible at this distance (13-20% of attempts, Table 1), but the picked branch spans a visibly narrower frequency range than quartiles 1-2, and low-quality crossings (top panel) are more frequent throughout the record.

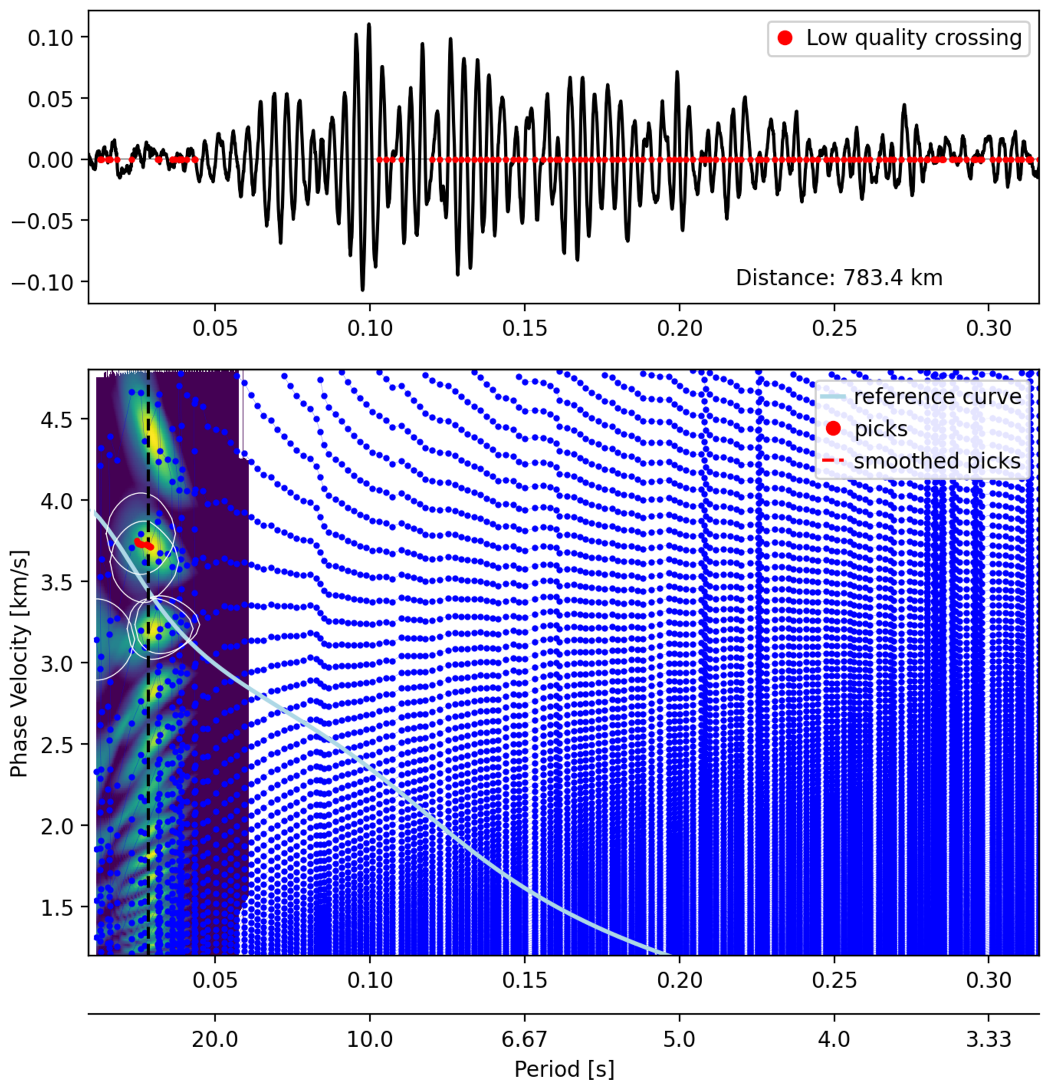

Figure 4: Quartile 4 (farthest), `XVDGOS_XVMAGY`, 783.4 km, $NW = 7.61$ ($1.5 \times NW_{\text{high}} = 5.08$; $W \approx 0.000704$ Hz), did not converge – the mechanism behind H3's finding, visually, within the picker's own cropped view. Over this narrower window the coherence spectrum (top) shows no clean, decaying envelope, and the majority of candidate crossings across the low-frequency half are flagged low-quality (the near-continuous wall of red). The KDE panel (bottom) shows only a thin sliver of real density at the extreme long-period edge; the picker finds a few points (small red cluster) and then has nothing left to track. **Correction, Section 8.2**: over the full 0–0.35 Hz band this pair's coherence in fact has a real, strong, multi-lobed envelope – it is not simply unstructured noise; the picker's own plot window (cropped near the low-frequency edge) does not show it.

`diff(w_axis) > 1.5*expected_fstep`) both use this quantity; only criterion 1 (the peak-ratio test) is reference-curve-independent. If the reference curve's velocity is wrong for a specific path, `expected_fstep` is wrong, and genuinely real, well-formed crossings can be flagged low-quality simply because their spacing does not match a miscalibrated prediction – not because the crossing itself is spurious.

This is not yet quantified for the dataset as a whole. What can be said now: this is a real, code-confirmed mechanism, not a hypothesis about the picker's behavior, and it means the `bad_quality_fraction` diagnostic reported throughout this document (and Round 1's own manifest) is not a pure measure of coherence quality – it is a joint measure of coherence quality *and* reference-curve accuracy for that specific path, and the two are not currently separated. A proper follow-up (not done here) would compare `bad_quality_fraction` against an independent measure of how far the reference curve sits from the picked/high-density branch, per pair, to check how much of the quality-gate signal is actually reference-curve error rather than data quality.

A local, synthetic isolation test. To make the mechanism concrete rather than just citing the formula, a fully local test (synthetic flat-velocity Bessel coherence, 500 km, no real station data needed) fed the *same* 376 zero crossings through the *same* unmodified picker logic, changing only the reference curve. Criterion 1 (peak-ratio) was, as its formula predicts, completely unchanged (235 flagged in both runs). Criterion 2 (`crossamps<0.5*peakamps`) nearly doubled from a correct reference curve (3.5 km/s, matching the synthetic truth) to one offset by 1.5 km/s: 119 vs. 210 flagged, out of 376, with every other input held fixed. This isolates the effect cleanly: the same data, the same crossings, a different reference curve alone changes which crossings are called low-quality.

Three distinct roles, not one. The reference curve enters the pipeline in three separate places, worth distinguishing explicitly: (1) the quality gate above; (2) resolving the zero crossing's fundamental 2π cycle-count ambiguity – `get_zero_crossings` compares every observed crossing against every Bessel-function zero, producing one velocity candidate per possible cycle count, and the reference curve is what picks which candidate is plausible, both to seed the walk and in the local-slope continuity checks as picking proceeds; (3) it defines the frequency band the final `freq_coverage_fraction` diagnostic is even measured against (`f_min`, `f_max` are taken directly from `min/max(ref_curve[:,0])`). A reference-curve mismatch is therefore not just a labeling artifact on individual crossings – it can also shift which candidate branch the walk starts on and what frequency range "coverage" means for that pair.

Is the mismatch specific to quartile 4? No. Checked directly against Figures 1–3's reference curves (light blue), not assumed: in all three converged examples (quartiles 1–3), the reference curve starts high, descends steeply, and *crosses through* the real, high-density picked branch around period 0.04–0.11 s, then continues to fall well below the true branch for the rest of the record – the same shape and roughly the same magnitude of mismatch visible in quartile 4. Since quartiles 1–3 converge despite this and quartile 4 does not, the reference-curve mismatch by itself cannot be the primary driver of the near/far convergence gap; it is a real, consistently-present confound across every distance tested, not a quartile-4-specific one. Within the picker's own cropped diagnostic view, quartiles 1–3 show a clear decaying-envelope wavetrain and quartile 4 does not – but Section 8.2 shows this reading does not survive looking at the full band: quartile 4 does have a real, strong envelope, just positioned differently and more sharply multi-lobed than the picker's crop reveals. The reference-curve mechanism should still be quantified dataset-wide (it is a real source of noise in every `bad_quality_fraction` number in this report).

8.2 FastMspec vs. single-taper, and the full-band envelope

Raised directly: how much better are FastMspec's correlations than single-taper's, even after the seislib gate – independent of picking altogether? Figures 5–8 overlay the two techniques' raw coherence (real part) for the same four pairs, over the full 0–0.35 Hz band computed for each (wider than the picker's own cropped diagnostic plots above). In every quartile, the FastMspec curve (red) sits as a visibly smoothed envelope through the single-taper curve's noise (grey) – both trace the same real zero crossings, but single-taper's version is far noisier, with many more spurious small-amplitude crossings that don't reflect real structure. This is the gate-independent comparison Section 5's Synthesis had flagged as missing.

This view also corrects the earlier reading of quartile 4 (Section 8.1 and Figure 4's caption): over the full band, quartile 4's coherence has a real, strong, multi-lobed envelope, with its main burst around 0.09–0.13 Hz – not the unstructured noise the picker's own cropped plot suggested. Quartile 3 (Figure 7) shows a comparably multi-lobed, two-burst envelope and still converges, so multi-lobed structure by itself does not distinguish quartile 4 from the converged examples. What does still look distinctive, and is being tested directly as this document is being written (results not yet in hand): quartile 4's strongest envelope lobe sits well inside the band rather than at the low-frequency edge, while the picker's walk (`extract_dispcurve`'s outer loop) always starts at the lowest-frequency non-`bad_quality` crossing and only ever advances upward in frequency – it has no mechanism to seed at, or expand outward from, the record's strongest region. If quartile 4's opening stretch is disproportionately unreliable relative to quartiles 1–3's, the walk may simply never reach its own strongest structure before stalling. A direct test – raising `freqmin` per pair to the frequency of peak local amplitude within the low-frequency half of the band, then rerunning the unmodified picker – is in progress; this section will be updated with the result.

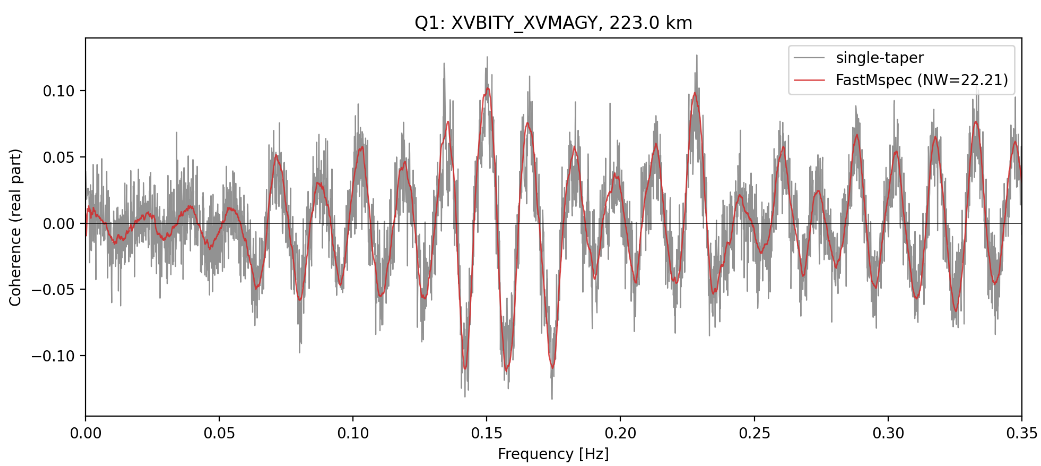

Figure 5: Quartile 1, FastMspec ($NW = 22.21$) vs. single-taper, full band.

9 Synthesis and Implications for Stage 5

- **H1 and H3 together give a clean, load-bearing result**: normalized bandwidth ($NW/NW_{\text{high}}(r)$) is the right variable for pairs where a viable window exists at all (quartiles 1–3), and the existence-threshold theory correctly predicts *which* pairs don't have one (quartile 4, confirmed with zero exceptions). This validates the core machinery `round2_prep.py` was

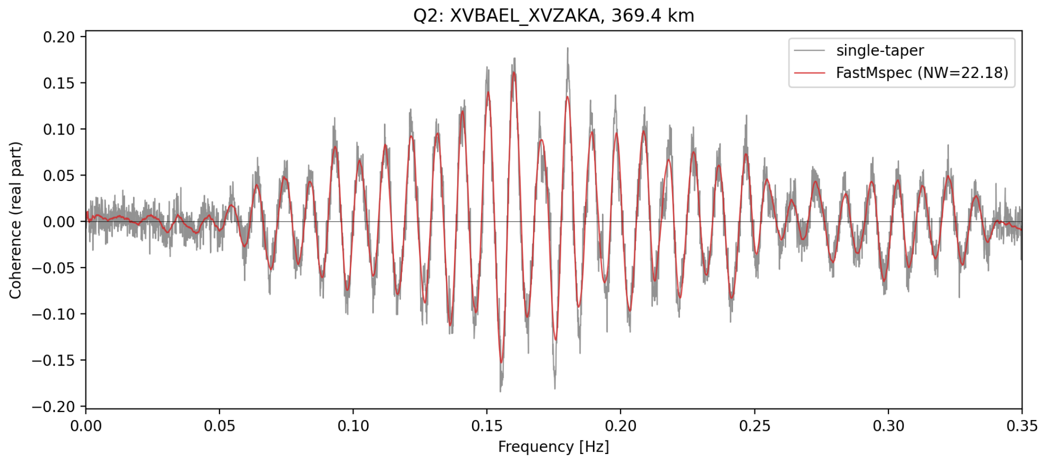

Figure 6: Quartile 2, FastMspec ($NW = 22.18$) vs. single-taper, full band.

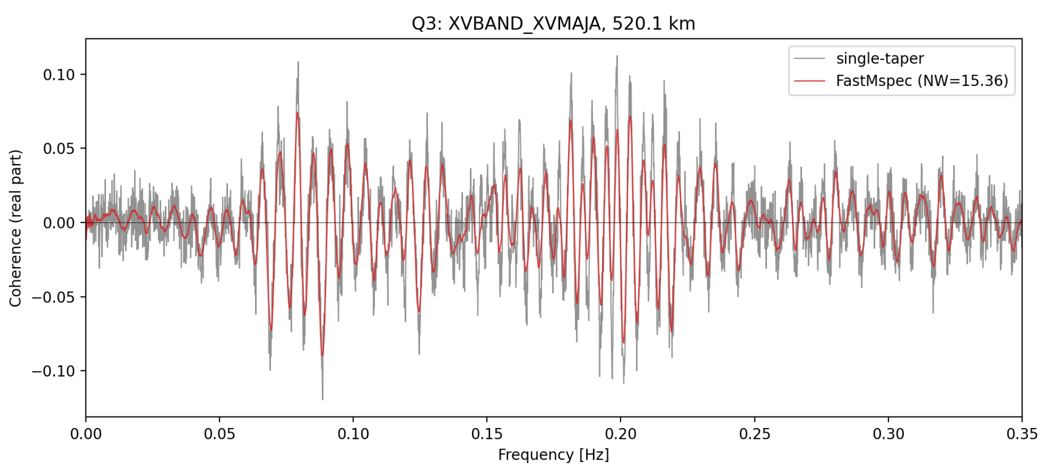

Figure 7: Quartile 3, FastMspec ($NW = 15.36$) vs. single-taper, full band. A comparably multi-lobed envelope to quartile 4, yet this pair converges – multi-lobed structure alone does not explain the difference.

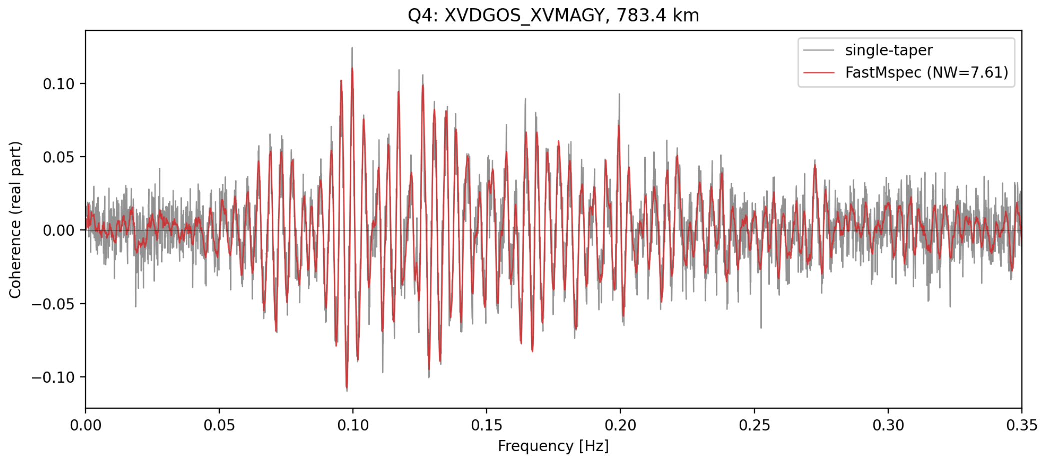

Figure 8: Quartile 4, FastMspec ($NW = 7.61$) vs. single-taper, full band. A real, strong envelope is present, peaking around 0.09–0.13 Hz – corrects the impression from Figure 4's cropped view.

built on, not just the sweep's raw convergence numbers.

- **H2's non-result is the most actionable finding for Stage 5's next step**: the bias–variance MSE framework predicts an interior optimum, and this sweep didn't find one – not because it was falsified, but because the tested range may not have reached it. Before trusting `stage5_bandwidth_theory.tex`'s Eq. 16 recipe as a complete answer, extend the sweep's upper fraction (e.g. to 2.0–3.0 $\times NW_{\mathrm{high}}$) on a subset of quartile-1 pairs specifically, where a real signal is most likely to be visible against the noise floor.
- **H4 is now a known, permanent constraint on any future sweep design**: any bandwidth search must stay above $NW \approx 3$ or expect the same clean failure mode – already fixed to fail informatively rather than silently, but still worth excluding from statistical fits by construction (e.g. floor the fraction grid's minimum well above wherever $NW \approx 3$ falls for the pairs being tested) rather than discovered again.
- **The reference-curve dependence (Section 8.1) affects how much this report's own convergence numbers can be trusted as pure coherence-quality measures**, and needs its own quantified check, not just the mechanism confirmed above. Checked across quartiles, not just quartile 4: the mismatch is present with similar shape and magnitude in every converged example too (Figures 1–3), so it is a dataset-wide confound on `bad_quality_fraction`, not a quartile-4-specific artifact – and correspondingly it does not by itself explain *why* quartile 4 fails while quartiles 1–3 succeed. The top-panel amplitude envelope (present in quartiles 1–3, absent in quartile 4) remains the more likely primary differentiator, consistent with H3.
- **A direct FastMspec-vs-single-taper coherence comparison, independent of the seislib quality gate, is a real and currently missing check**, raised directly. Every convergence number in this report and in Round 1 (FastMspec 26.1% vs. single-taper 0%) is filtered through the picker's own gate – and that gate is now known (above) to be sensitive to reference-curve accuracy, not solely to the underlying coherence's own quality. Whether FastMspec's real advantage over single-taper is larger, smaller, or the same once measured directly on the coherence itself (e.g. an SNR or spectral-flatness metric computed before any picking), rather than through the gate's pass/fail outcome, is not yet known and should be checked before leaning further on the gated convergence rate as the primary comparison metric.

10 Caveats

- Sample sizes are modest – 15 pairs per quartile, and after excluding the $K = 0$ floor, as few as 22 valid units at the lowest fraction. The quartile-level patterns (Tables 1, 5) are stark enough to trust qualitatively; finer claims (e.g. exact convergence-rate percentages) carry real sampling uncertainty at this n.
- Quality scalars (Table 2, bottom two rows) are averaged only over *converged* units at each fraction – as convergence rate drops, this average is increasingly selected on a shrinking, potentially biased subset (as few as 3 units at fraction 0.25), not a random sample of "quality regardless of outcome."
- This report covers FastMspec only, the technique the sweep was designed around; single-taper's own Round 1 result (0/380 converged, a real mechanistic result) is separate context, not re-evaluated here.

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